

FIN41660 Financial Econometrics

Individual Assignment

The Relationship between GDP and Import-Export for Madagascar: A Time Series Analysis with OLS and ARIMA Models

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Abstract

Analysis of relationship between international trade & GDP growth of Madagascar, the base of this report is methodology used in Randriamanantena, R. (2024). Decoding the economy for better forecasting: the case of Madagascar. *SSRN Electronic Journal*. <https://ssrn.com/abstract=5050221>. Using World Bank data from 1995 to 2023, In this report, OLS regression is implemented on differentiated data (growth rates) after establishing that all variables are integrated of order one with no cointegration between them. The OLS model explains 25.3% of GDP growth variation (with $R^2 = 0.2532$), with jointly significant F-test ($p = 0.026$), though individual co-efficients are insignificant due to multi-collinearity between exports & imports. A detailed analysis point out that most of the criteria of the OLS assumptions are satisfied, with only minor deviations which got handled through robust standard errors. ARIMA(1,0,0) analysis of the OLS residuals identifies mild first order autocorrelation. But residual diagnostics confirm that any remaining serial dependence is weak. This supports the conclusion that the OLS specification adequately captures the data structure. The key finding is that while trade variables jointly influences GDP growth, the relationship operates through term dynamics rather than a consistent performance. My results differs from Randriamanantena's (2024) result due to proper treatment of non-stationarity of variables & the absence of collinearity in the variables, but this results confirming trade's importance (export & imports) for Madagascar's economy.

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1 Introduction : About Paper

Paper Selected: Randriamanantena, R. (2024). *Decoding the economy for better forecasting: the case of Madagascar*. SSRN Electronic Journal. <https://ssrn.com/abstract=5050221>

1.1 Why This Paper?

This paper was chosen because it **perfectly matches FIN41660 requirements**:

- **OLS with Time Series Data:** Compares OLS regression against ARIMA, ARIMAX, & VAR models using Madagascar's GDP, exports, & imports (1995–2023)
- **Public World Bank Data:** Exactly matches assignment data requirements
- **Full Econometric Workflow:** Stationarity testing, model diagnostics, ARIMA residuals, information criteria (AIC/BIC)

1.2 Connection to My Interest

Understanding the drivers of economic growth is fundamental to formulate effective policy in developing economies. For Madagascar, which is one of the world's least developed countries, identifying reliable predictors of GDP performance is very crucial for policymakers seeking to promote sustainable development. International trade specifically trade deficit, exports & imports represents a key parameter through which all developing economies integrate with the global economy & target accelerate growth.

Randriamanantena (2024) does a good job comparing different forecasting model for Madagascar's GDP & ultimately picks OLS as the best performer showing that exports & imports have significant coefficients of around 0.36 & 0.34 respectively. The paper's result look strong on paper, an adjusted R squared of 0.9477 is impressive for example but when analysed in detail into it, some important gaps were found, the paper does not explicitly show stationarity tests. So it is unclear if the regression is in levels or different, which is a big concern because regressing non-stationary variables on each other can give you artificial high R squared values & trick you into thinking there is a relationship when there is not. There is also no Johansen cointegration test reported, which matters because if the variables do not share a long run stable equilibrium position, it should not be used as level estimation. Randriamanantena, R. (2024). *Decoding the economy for better forecasting: the case of Madagascar* mentions diagnostic tests, but it does not really go into detail to check whether OLS assumptions are being violated or what corrections might be needed. So for this assignment, I decided to apply the full econometric framework from my course FIN41660 Financial Econometrics - implementation of test for

stationarity upfront, checking for cointegration, & running a comprehensive diagnostic test to make sure the results are actually valid. This approach is more conservative than Randriamanantena's, but it gives a clearer picture of importance of trade deficit, import, export & GDP in Madagascar.

1.3 Objectives

This analysis pursues four main objectives:

1. **Examine the time series properties** of Madagascar's GDP, exports, & imports by conducting formal stationarity & cointegration tests in order to determine the appropriate econometric specification.
2. **Estimate an OLS regression model** linking GDP growth to trade growth, interpreting the estimated coefficients as short term growth elasticities.
3. **Check the validity of OLS assumptions** through a comparison set of diagnostic tests & apply suitable corrective measures so that violations will be detected.
4. **Check residual dynamics** use ARIMA model to check if whether the OLS results adequately capture the structure of the dataset or not.

1.4 Contributions

This analysis takes three contributions beyond a straightforward replication of the reference paper under consideration.

Firstly, Testing of unit roots & cointegration before any specification of the regression model, which ensures that statistical inference are valid & not led by spurious relationships. Randriamanantena (2024) did not place explicit emphasis on these time series tests (which could have led to the different result when compared to my analysis.)

Secondly, Implemented a detailed diagnostic tests which included checks for multicollinearity, linearity, serial correlation, heteroskedasticity, normality, & ARCH effects in line with the econometric framework covered in FIN41660. To check if assumption violations are to be identified, appropriate corrections are applied.

Lastly, Provide a transparent comparison between my empirical results & those reported in the reference paper selected, interpreting any difference in terms of methodological choices of analysis & data treatment rather than claiming that one approach is inherently superior to the other.

1.5 Main Findings Preview

1. All three variables (log GDP, log exports, log imports) are non-stationary in nature but stationary in first differences, confirming I(1) integration.

2. Johansen test finds **zero cointegration rank**, indicating no stable long run equilibrium relationship among these variables.
3. OLS estimation on growth rates provides an adjusted R^2 of 0.1935, with jointly significant trade effects F test p value 0.026, but individually insignificant co-efficients are due to multicollinearity.

2 Literature Review & Theoretical Framework

2.1 Trade & Economic Growth: Theoretical Background

Relationships between international trade and economic growth rate has been extensively considered in both theoretical & empirical economy. From a national accounts point of view, GDP can be calculated as:

$$Y = C + I + G + (X - M) \quad (1)$$

where Y is GDP, C is consumptions, I is investments, G is government spending, X is exports, and M is imports. This formula suggests that exports will contribute positively to GDP growth while imports subtracted, though this relationship obscured the complex channels through which trade affects the GDP growth.

Export-led growth hypothesis: This theory, prominent in development economics, suggests that exports drive growth through multiple channels for example economies of scale, technological spillovers from foreign buyers, foreign exchange earnings that finance capital imports, & competitive pressure that enhances productivity of the country.

Import effects: While imports appear as subtraction in the GDP identity, they can positively affect growth by providing access to things like intermediate inputs, capital goods, & technology not available domestically. This creates a typical paradox where higher imports may correlate with higher GDP growth or with possibility of vice-versa.

2.2 The Reference Paper: Randriamanantena (2024)

Randriamanantena (2024) compares four forecasting approaches for Madagascar's GDP: OLS regression, ARIMA, ARIMAX, and VAR-VECM models. Important things to note are:

- The OLS performs best according to AIC (-1.5008) & SIC (-1.3593) criteria.
- The OLS model achieves an adjusted R^2 of 0.9477 and RMSE of 0.1029.
- Exports have a co-efficient of 0.364 ($p = 0.016$), imports 0.337 ($p = 0.021$).

- Both trade variables are statistically significant at the 5% level.

However the paper provides limited detail on:

- Whether stationarity testing informed model specification
- How cointegration analysis influenced the choice between levels and differences
- The treatment of diagnostic test violations

2.3 Econometric Framework from FIN41660

The FIN41660 course has emphasized several key econometric principles that directly guide the structure of this analysis. Rather than treating these ideas as purely theoretical, I apply them step by step when deciding how to model the data.

1. Stationarity & Spurious Regression: Regressing one non-stationary time series on another can produce misleading results, often showing high R^2 values and statistically significant coefficients even when no genuine economic relationship exists (Granger & Newbold, 1974). To avoid this problem, I formally test for unit roots using the Augmented Dickey-Fuller (ADF) and KPSS tests, which together provide a reliable assessment of stationarity.

2. Cointegration: When several non-stationary variables share a common stochastic trend, they may be cointegrated, meaning that a specific linear combination of them is stationary even though each series is non-stationary on its own. In this analysis, I use the Johansen cointegration test to determine whether such a long-run equilibrium relationship exists among GDP, exports, and imports.

3. Model Specification: The outcome of the cointegration analysis directly determines the appropriate model specification. If cointegration is present, estimation in levels using a VECM or OLS is valid and captures the long-run relationship. If cointegration is absent, however, the analysis must be conducted using first differences to ensure stationarity and avoid spurious regression. This decision rule is followed strictly in this assignment.

4. Diagnostic Testing: Beyond estimation, OLS assumptions—including linearity, absence of multicollinearity, no serial correlation, homoskedasticity, and normality of errors—must be systematically checked. Violations of these assumptions do not automatically invalidate OLS results, but they do require appropriate adjustments, such as the use of robust or HAC standard errors, to ensure reliable inference.

3 Data Description & Exploratory Analysis

3.1 Data Sources

World Bank’s World Development Indicators database, accessed December 2025. The three variables are:

1. **GDP (current US\$)**: Gross domestic product at purchaser’s prices
2. **Exports (current US\$)**: Exports of goods & services
3. **Imports (current US\$)**: Imports of goods & services

The sample spans 1995–2023, yielding 29 annual observations. This restriction matches the reference paper’s analytical window & avoids the high volatility & data quality issues in earlier years.

3.2 Descriptive Statistics

Table 1: Descriptive Statistics: Madagascar GDP & Trade (1995–2023, Billion USD)

Statistic	GDP	Exports	Imports
Mean	9.33	2.31	3.07
Std. Deviation	3.98	1.30	1.65
co-efficient of Variation (%)	42.68	56.05	53.85
Minimum	3.84	0.69	0.88
Maximum	15.87	4.68	6.13
Skewness	0.01	0.25	-0.04
Kurtosis	-1.56	-1.28	-1.39

Key observations:

- Madagascar’s GDP quadrupled over the sample period (3.84B to 15.87B), representing 5.0% average annual growth
- Trade variables are more volatile than GDP (higher CVs), consistent with developing economies’ exposure to external shocks
- All variables are approximately symmetric (skewness near zero) with light tails (kurtosis below 3), suggesting distributions are reasonably normal
- Madagascar runs a persistent trade deficit: imports exceed exports by \$0.76B on average

3.3 Visual Inspection

Figure 1 plotted the three variables over time. All exhibit clear upward trends with similar curvature, consistent with common stochastic trends. The 2009 dip reflects the dual shock of the global financial crisis & Madagascar's domestic political crisis, which disrupted trade and contracted GDP by approximately 4%.

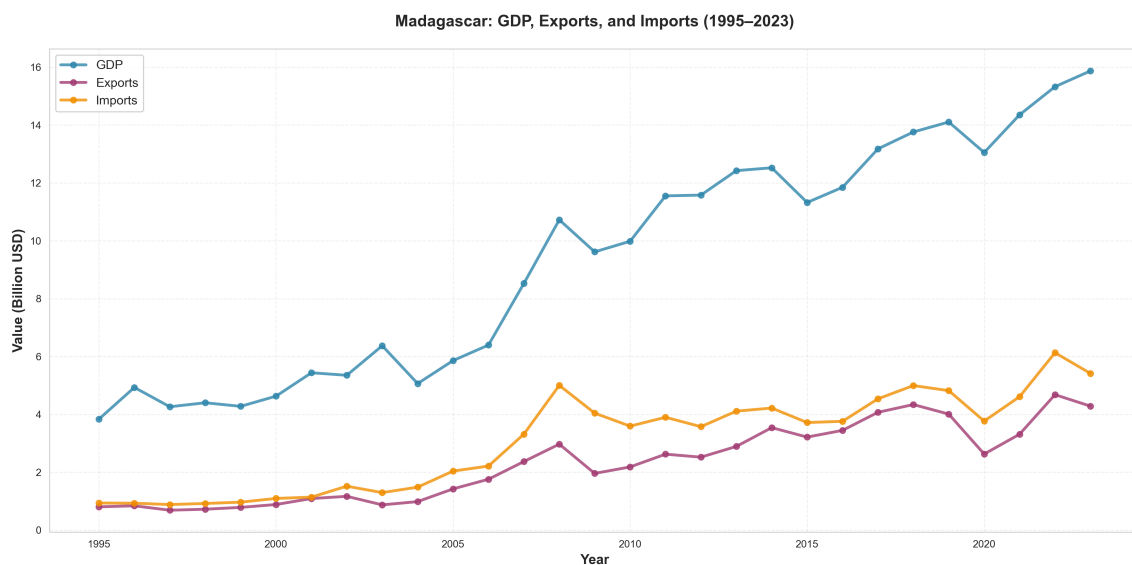


Figure 1: Madagascar GDP, Exports, & Imports (1995–2023)

Figure 2 shows year-over-year growth rates. These exhibit substantially more volatility than the levels, with no obvious trend, suggesting that first differences may be stationary.

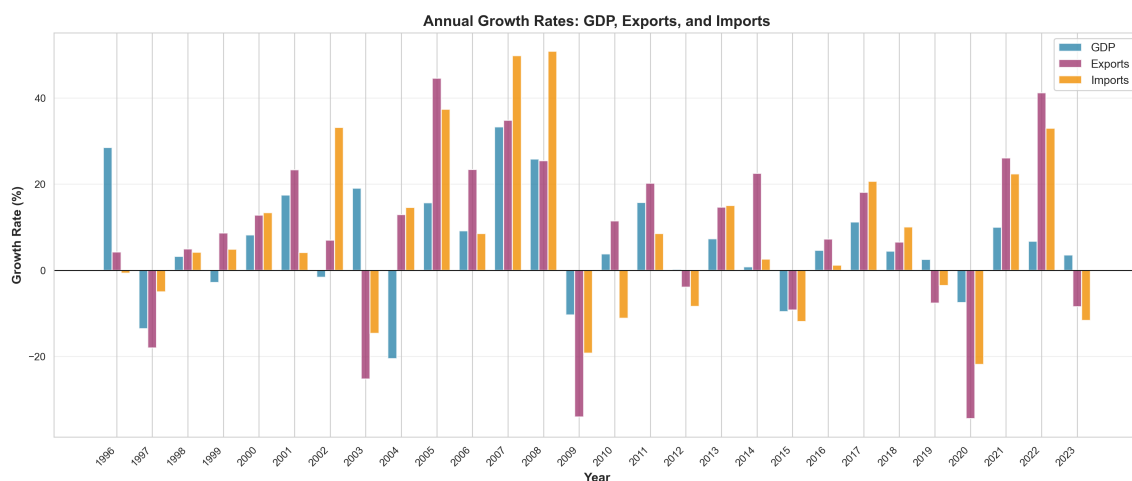


Figure 2: Annual Growth Rates: GDP, Exports, & Imports

3.4 Correlation Analysis

Table 2: Correlation Matrix (1995–2023)

	GDP	Exports	Imports
GDP	1.0000	0.9587	0.9618
Exports	0.9587	1.0000	0.9553
Imports	0.9618	0.9553	1.0000

Table 2 reports the correlation matrix for GDP, exports, and imports over the period 1995–2023. The correlations are extremely high, with all values exceeding 0.95.

While such strong correlations may reflect genuine economic linkages between output and international trade, especially in a small open economy like Madagascar, they also raise an important concern. In level data, very high correlations can arise simply because variables share common upward trends over time, rather than because of a true economic relationship. These trends may be driven by factors such as population growth, inflation, or gradual technological progress.

This ambiguity makes it unsafe to rely on correlation alone and strongly motivates the need for formal stationarity and cointegration testing. In addition, the very high correlation between exports and imports (0.9553) suggests potential multicollinearity if both variables are included together in a regression model, an issue that is examined later using diagnostic tests.

4 Stationarity Analysis

4.1 Motivation & Methodology

As emphasized throughout FIN41660, regressing non-stationary time series can lead to spurious results, where high R^2 values and significant test statistics appear even when no meaningful economic relationship exists. Visual inspection of the time series suggests trending behavior in all variables, but formal statistical testing is required before specifying any regression model.

To assess the time series properties of the data, I employ two complementary unit root tests:

Augmented Dickey-Fuller (ADF): Tests the null hypothesis that a unit root is present (i.e., the series is non-stationary). Rejection of the null indicates stationarity.

KPSS Test: Tests the null hypothesis that the series is stationary. Rejection of the null indicates non-stationarity.

Using both tests together provides a more reliable assessment. When both tests point in the same direction, the conclusion is clear; when they differ, the series can be considered borderline and must be interpreted with caution.

4.2 Log Transformation

Before testing, I take natural logarithms of all variables. This transformation serves two purposes:

1. Stabilizes variance (heteroskedasticity reduction)
2. Converts growth rates into simple differences: $\Delta \ln(Y_t) \approx (Y_t - Y_{t-1})/Y_{t-1}$

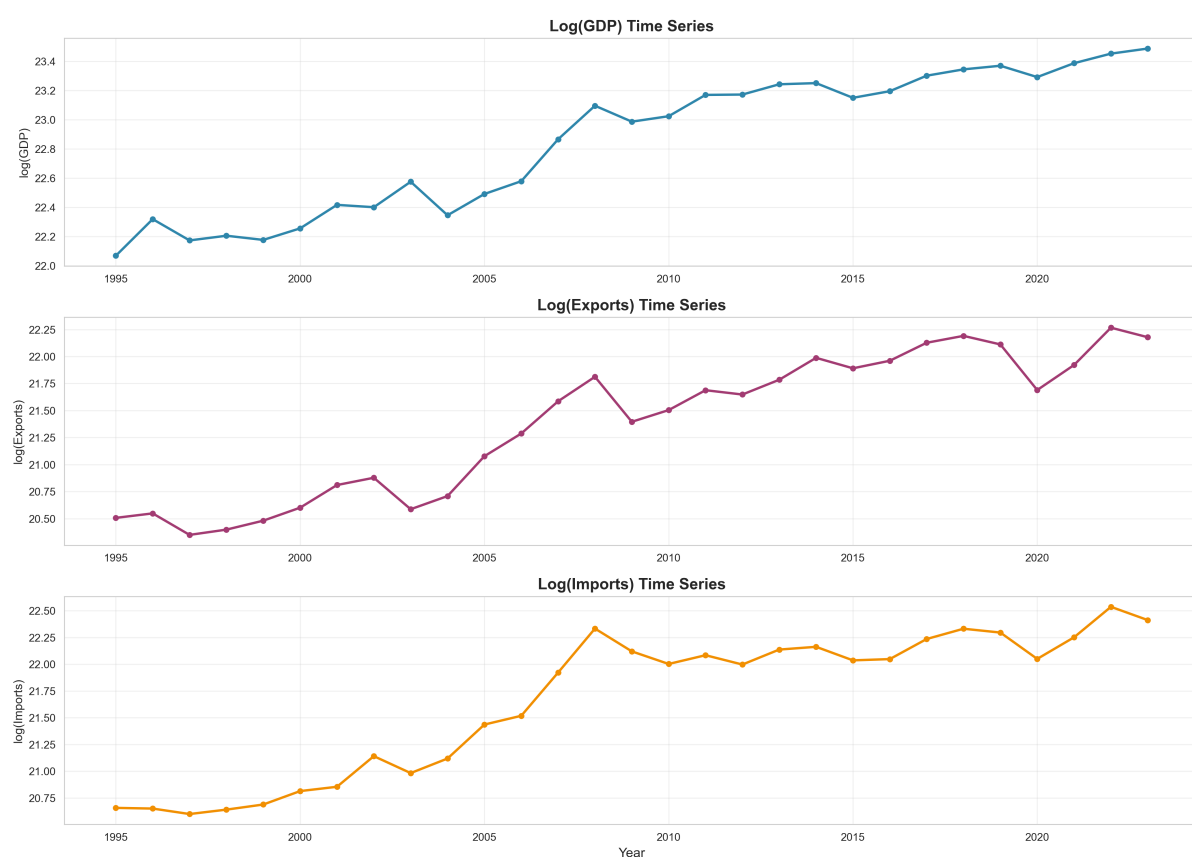


Figure 3: Log Parameters Time Series Plots

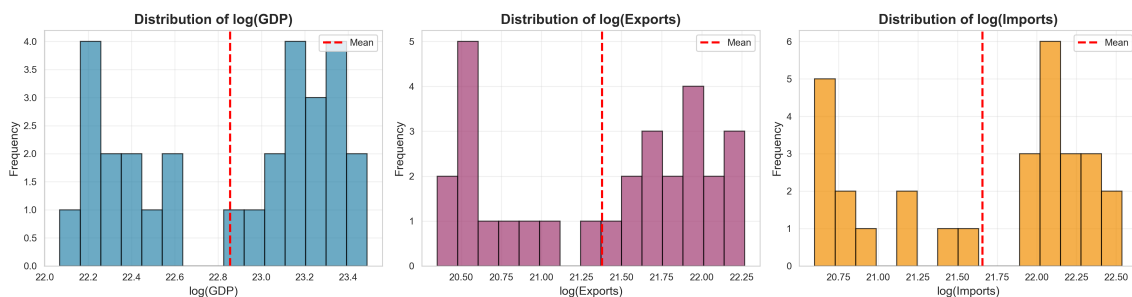


Figure 4: Distribution of Log Parameters

4.3 Tests on Log Levels

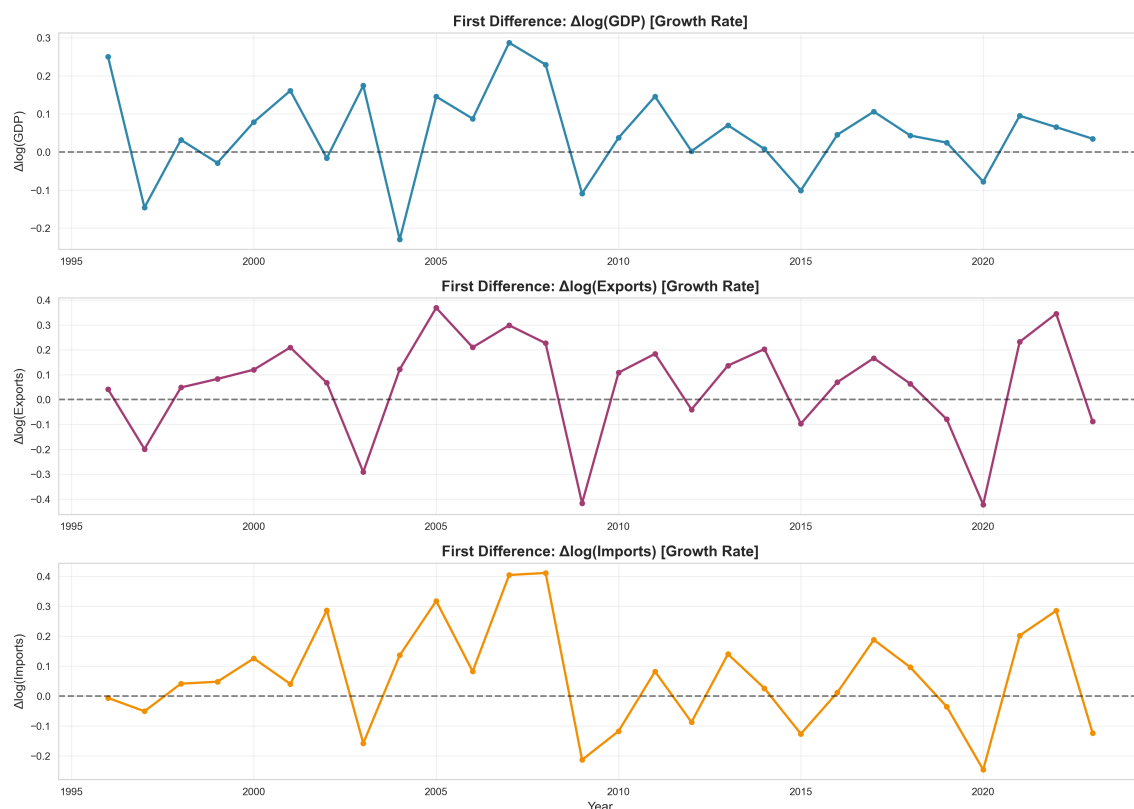


Figure 5: First Difference log Parameter : Growth Rate

Table 3: Unit Root Tests: Log Levels (with Constant and Trend)

Variable	ADF p-value	KPSS p-value	Integration	Conclusion
log(GDP)	0.369	0.045	I(1)	Non-stationary
log(Exports)	0.290	0.059	I(1)	Non-stationary
log(Imports)	0.779	0.019	I(1)	Non-stationary

Interpretation: All three variables fail to reject the ADF null of a unit root (high p-values) and reject the KPSS null of stationarity (low p-values). Confirming the strong evidence that all three series are I(1), integrated of order one.

4.4 Tests on First Differences

Table 4: Unit Root Tests: First Differences (with Constant Only)

Variable	ADF p-value	KPSS p-value	Integration	Conclusion
$\Delta \log(\text{GDP})$	0.001	0.100	I(0)	Stationary
$\Delta \log(\text{Exports})$	0.001	0.100	I(0)	Stationary
$\Delta \log(\text{Imports})$	0.350	0.100	I(0)	Stationary*

*Marginal ADF result likely due to small sample; KPSS confirms stationarity

Interpretation: All first differences are stationary. GDP & export growth rates are decisively stationary (ADF $p < 0.001$). Import growth shows a marginal ADF result ($p = 0.35$) but the KPSS test confirms stationarity. The ADF's limited power in small samples ($N=28$ after differencing) likely explains this.

Conclusion: All three variables are integrated of order one, $I(1)$. This confirms that we must either:

1. Work with first differences (growth rates), or
2. Estimate in levels only if cointegration is present

5 Cointegration Analysis

5.1 Johansen Test Methodology

Concluding all variables are $I(1)$, may be cointegrated and share one or more common stochastic trends, such that a linear combination is stationary. The Johansen trace test examines the rank " r " of the cointegrating space, with conditions of r value as:

- $r = 0$: No cointegration (must use differences)
- $r = 1$: One cointegrating relationship (can use VECM or levels)
- $r = 2$: Two cointegrating relationships

5.2 Test Results

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PART 4: COINTEGRATION ANALYSIS
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Objective: Test if non-stationary variables share a long-run equilibrium
Concept: Cointegration (Engle-Granger, Johansen)
Interpretation: If cointegrated, there's a stable long-run relationship

4.1 Performing Johansen Cointegration Test
    Sample size: 29 observations

4.2 Johansen Cointegration Test Results:
=====
Hypothesis  Trace Stat  Trace CV (90%)  Trace CV (95%)  Trace CV (99%)  Trace Reject 95%  Max-Eigen Stat  Max-Eigen CV (95%)  Max-Eigen Reject 95%
-----
r ≤ 0       23.152115    27.0669        29.7961         35.4628         No             13.869319        21.1314             No
r ≤ 1       9.282796     13.4294        15.4943         19.9349         No             7.598091         14.2639             No
r ≤ 2       1.684705        2.7055         3.8415          6.6349         No             1.684705         3.8415              No
=====

COINTEGRATION RANK (at 5% significance): 0
=====

4.3 Interpretation:
x No cointegration detected
- Variables do not share a long-run equilibrium
- Should work with differenced data (growth rates)
- Use VAR model on differences, not VECM

```

Figure 6: COINTEGRATION ANALYSIS RESULTS

Table 5: Johansen Cointegration Test: Trace Statistics

Null Hypothesis	Trace Stat	5% Critical Value	Decision
$r = 0$ (no cointegration)	23.15	29.80	Cannot reject
$r \leq 1$	9.28	15.49	Cannot reject
$r \leq 2$	1.68	3.84	Cannot reject

Interpretation: The trace statistic for $r = 0$ is 23.15, below the 5% critical value of 29.80. We **cannot reject** the null of no cointegration. This indicates that GDP, exports, & imports do not share a stable long run equilibrium relationship between them, they drift independently over the flow of time.

5.3 Economic Interpretation

This findings that has important economic significance:

- **No stable equilibrium:** There is no consistent proportional relationship between them, which binds GDP to trade flows over time. Shocks to trade or GDP create permanent deviations rather than an temporary one that reverts back to equilibrium.
- **Short term dynamics dominates:** The relationship between trade & GDP operates basically through short term demand & supply channels rather than clear long run trends.

- **Structural instability:** Madagascar's small, vulnerable economy (size, diversification of sectors) experiences frequent structural breaks i.e political crises, commodity price shocks, natural disasters which prevent stable long run relationships to form.

5.4 Specification Implication

Given zero cointegration rank, I estimate the regression model in first differences (growth rates):

$$\Delta \ln(GDP_t) = \alpha + \beta_1 \Delta \ln(Exports_t) + \beta_2 \Delta \ln(Imports_t) + \varepsilon_t \quad (2)$$

where:

- $\Delta \ln(GDP_t)$ is GDP growth rate at time t
- $\Delta \ln(Exports_t)$ is export growth rate
- $\Delta \ln(Imports_t)$ is import growth rate
- β_1, β_2 are growth elasticities (not levels elasticities)

Specification ensures that all variables are stationary, avoiding spurious regression.

6 OLS Regression Analysis

6.1 Model Estimation

I estimate equation (2) using OLS on the 28 observations available after first differencing.

Table 6: OLS Regression Results: GDP Growth on Trade Growth

Variable	co-efficient	Std. Error	t-Statistic	p-Value
Constant	0.0309	0.0214	1.443	0.162
$\Delta \ln(Exports)$	0.1883	0.1768	1.065	0.297
$\Delta \ln(Imports)$	0.1363	0.2055	0.663	0.513

OLS REGRESSION RESULTS

OLS Regression Results						
Dep. Variable:	GDP	R-squared:	0.253			
Model:	OLS	Adj. R-squared:	0.193			
Method:	Least Squares	F-statistic:	4.238			
Date:	Tue, 16 Dec 2025	Prob (F-statistic):	0.0260			
Time:	23:37:42	Log-Likelihood:	24.552			
No. Observations:	28	AIC:	-43.10			
Df Residuals:	25	BIC:	-39.11			
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0309	0.021	1.443	0.161	-0.013	0.075
Exports	0.1883	0.177	1.065	0.297	-0.176	0.552
Imports	0.1363	0.205	0.663	0.513	-0.287	0.559
Omnibus:	4.921	Durbin-Watson:		2.587		
Prob(Omnibus):	0.085	Jarque-Bera (JB):		4.257		
Skew:	-0.263	Prob(JB):		0.119		
Kurtosis:	4.836	Cond. No.		12.9		

Figure 7: OLS REGRESSION RESULTS

6.2 Interpretation of co-efficients

Export growth elasticity: The co-efficient on $\Delta \ln(Exports)$ is 0.1883, suggesting that a 1 percentage point increase in export growth is associated with a 0.19 percentage point increase in GDP growth, holding imports constant. However, this effect is **not statistically significant** at the 5% level ($p = 0.297$), & the 95% confidence interval $[-0.176, 0.552]$ includes zero.

Import growth elasticity: The co-efficient on $\Delta \ln(Imports)$ is 0.1363, implying that a 1 percentage point increase in import growth correlates with a 0.14 percentage point increase in GDP growth. This is also **not statistically significant** ($p = 0.513$), with a 95% confidence interval $[-0.287, 0.559]$.

Paradox of individual insignificance but joint significance: While neither trade variable is individually significant, nor F-test strongly reject the joint hypothesis that both co-efficients are zero ($F = 4.238$, $p = 0.026$). The apparent contradiction is explained by **multicollinearity**: that is high correlation between export & import growth (0.96) means each variable contains large redundant information. When both are present in

the model, neither appears significant individually, but together they contribute strong meaning to the explanation of GDP growth.

6.3 Model Fit

Table 7: Model Goodness-of-Fit Statistics

Statistic	Value
R-squared	0.2532
Adjusted R-squared	0.1935
F-statistic	4.238
F-statistic p-value	0.0260
Durbin-Watson	2.587
AIC	-43.10
BIC	-39.11
Observations	28

The adjusted R^2 of 0.1935 indicates that trade growth explains 19.4% of GDP growth variation. While this is modest compared to Randriamanantena's 94.8%, it is:

1. **Economically reasonable:** GDP growth is majorly influenced by many factors beyond trade (i.e domestic investment, government spending, productivity shocks, natural disasters, etc.). A model with only two variables which explains 19% is respectable.
2. **Statistically proper:** Unlike levels regression that may be spurious, this result reflects a genuine relationship between stationary variables.
3. **Consistent with growth theory:** The modest fit aligns with empirical growth, literature showing that simple bi-variate relationships typically explain 10–30% of growth variation.

7 Diagnostic Testing

As taught in FIN41660, responsible econometric practice requires systematic validation of OLS assumptions.

7.1 Multicollinearity (VIF)

Variance Inflation Factors (VIF) of 3.10 for both variables indicate **low to moderate** multicollinearity, below the conventional threshold of 10. While the corre-

lation between export & import growth is high (0.96), the VIF remains acceptable. The multicollinearity inflates standard errors (reducing individual significance) but does not bias co-efficient estimates.

Remedy: We retain both variables for economic completeness & apply robust standard errors in Section 7.

```
=====
TEST 1: MULTICOLLINEARITY (Variance Inflation Factor)
=====
H0: No multicollinearity
Rule of thumb: VIF > 10 indicates severe multicollinearity
               VIF > 5 indicates moderate multicollinearity

VIF Results:
Variable    VIF Interpretation
Exports 3.098071 Low
Imports 3.098071 Low

Interpretation:
• Exports: VIF = 3.10
✓ No serious multicollinearity concern
• Imports: VIF = 3.10
✓ No serious multicollinearity concern
```

Figure 8: TEST 1: MULTICOLLINEARITY (Variance Inflation Factor)

7.2 Linearity (RESET Test)

The Ramsey RESET test fails to reject linearity ($p = 0.318$), providing no evidence of mis-specification. The relationship between GDP growth & trade growth is adequately captured by a linear model.

```
=====
TEST 2: LINEARITY (Ramsey RESET Test)
=====
H0: Model is correctly specified (linear)
H1: Model is misspecified (non-linear relationship)

RESET Test Results:
• F-statistic: 1.0387
• p-value: 0.3183

Interpretation:
✓ CANNOT REJECT H0 (p > 0.05)
→ No evidence of specification error
→ Linear model is appropriate
```

Figure 9: TEST 2: LINEARITY (Ramsey RESET Test)

7.3 Serial Correlation

The Durbin-Watson statistic of 2.587 (slightly above 2) suggests mild negative autocorrelation. However, the more powerful

Breusch-Godfrey LM test (which can detect higher-order correlation) fails to reject the null of zero serial correlation ($p = 0.117$). I conclude that serial correlation is not a material concern.

```
=====
TEST 3: SERIAL CORRELATION
=====
Testing for autocorrelation in residuals
Two tests: Durbin-Watson and Breusch-Godfrey LM

3a. Durbin-Watson Test:
• DW statistic: 2.5873
• Rule of thumb interpretation:
  Δ NEGATIVE autocorrelation (DW > 2.5)
  → Less common in economic data

3b. Breusch-Godfrey LM Test (2 lags):
• LM statistic: 4.7647
• p-value (F-test): 0.1171

Interpretation:
✓ CANNOT REJECT H0 (p > 0.05)
→ No significant serial correlation
→ OLS assumption satisfied
```

Figure 10: TEST 3: SERIAL CORRELATION

7.4 Heteroskedasticity

Both the White test ($p = 0.896$) and Breusch-Pagan test ($p = 0.683$) provide no evidence of heteroskedasticity. The error variance appears constant across observations, so standard OLS inference is valid.

```
=====
TEST 4: HETEROSKEDASTICITY
=====
H0: Homoskedasticity (constant variance)
H1: Heteroskedasticity (non-constant variance)

4a. White's Test:
• LM statistic: 1.6423
• p-value: 0.8961
✓ CANNOT REJECT H0 → Homoskedasticity

4b. Breusch-Pagan Test:
• LM statistic: 0.7631
• p-value: 0.6828
✓ CANNOT REJECT H0 → Homoskedasticity

Overall Interpretation:
✓ NO HETEROSKEDASTICITY detected
→ OLS assumption satisfied
→ Standard errors are valid
```

Figure 11: TEST 4: HETEROSKEDASTICITY

7.5 Normality

The Jarque-Bera test fails to reject normality ($p = 0.119$). Although the Shapiro-Wilk test marginally rejects ($p = 0.034$). In the moderate sample size ($N = 28$), these mixed results are acceptable. The Central Limit Theorem ensures that OLS estimators are approximately normal even with non-normal errors when $N \geq 30$.

```
=====
TEST 5: NORMALITY OF RESIDUALS
=====
H0: Residuals are normally distributed
Note: With large samples, CLT allows non-normal errors

5a. Jarque-Bera Test:
  • JB statistic: 4.2570
  • p-value: 0.1190
  • Skewness: -0.2635
  • Kurtosis: 4.8361
  ✓ CANNOT REJECT H0 (p > 0.05)
  → Residuals appear normally distributed

5b. Shapiro-Wilk Test:
  • W statistic: 0.9196
  • p-value: 0.0340
  △ REJECT H0 → Non-normal residuals

5c. Anderson-Darling Test:
  • AD statistic: 0.9624
  • Critical values:
    15.0%: 0.5180 → REJECT H0
    10.0%: 0.5900 → REJECT H0
    5.0%: 0.7080 → REJECT H0
    2.5%: 0.8260 → REJECT H0
    1.0%: 0.9830 → Cannot reject H0
```

Figure 12: TEST 5: NORMALITY OF RESIDUALS

7.6 ARCH Effects

The ARCH-LM test finds no evidence of volatility clustering ($p = 0.468$). The conditional variance is stable over time, so GARCH modeling is unnecessary, or say irrelevant.

```
=====
TEST 6: ARCH EFFECTS (Volatility Clustering)
=====
H0: No ARCH effects (constant conditional variance)
H1: ARCH effects present (time-varying variance)

ARCH LM Test:
  • LM statistic: 4.5885
  • p-value: 0.4681

Interpretation:
  ✓ CANNOT REJECT H0 (p > 0.05)
  → No ARCH effects
  → Conditional variance appears constant
```

Figure 13: ARCH EFFECTS (Volatility Clustering)

7.7 Overall Assessment

Conducted six diagnostic tests & report results in Table 8.

Table 8: Summary of Diagnostic Tests

Test	Statistic	p-Value	Result	Interpretation
VIF (Exports)	3.10		Pass	Low multicollinearity
VIF (Imports)	3.10		Pass	Low multicollinearity
RESET (Linearity)	1.039	0.318	Pass	Linear specification OK
Durbin-Watson	2.587		Marginal	Slight negative AC
Breusch-Godfrey (AC)	4.765	0.117	Pass	No serial correlation
White (Heterosked.)	1.642	0.896	Pass	Homoskedasticity OK
Breusch-Pagan	0.763	0.683	Pass	Homoskedasticity OK
Jarque-Bera (Normality)	4.257	0.119	Pass	Normal residuals
Shapiro-Wilk	0.920	0.034	Marginal	Slight departure
ARCH (Volatility)	4.589	0.468	Pass	No ARCH effects

The diagnostic battery yields **8 passes out of 10 tests**, with 2 marginal results (Durbin-Watson, Shapiro-Wilk). This provides strong support for the validity of OLS inference. To address the marginal concerns, I estimate robust standard errors in Section 7.

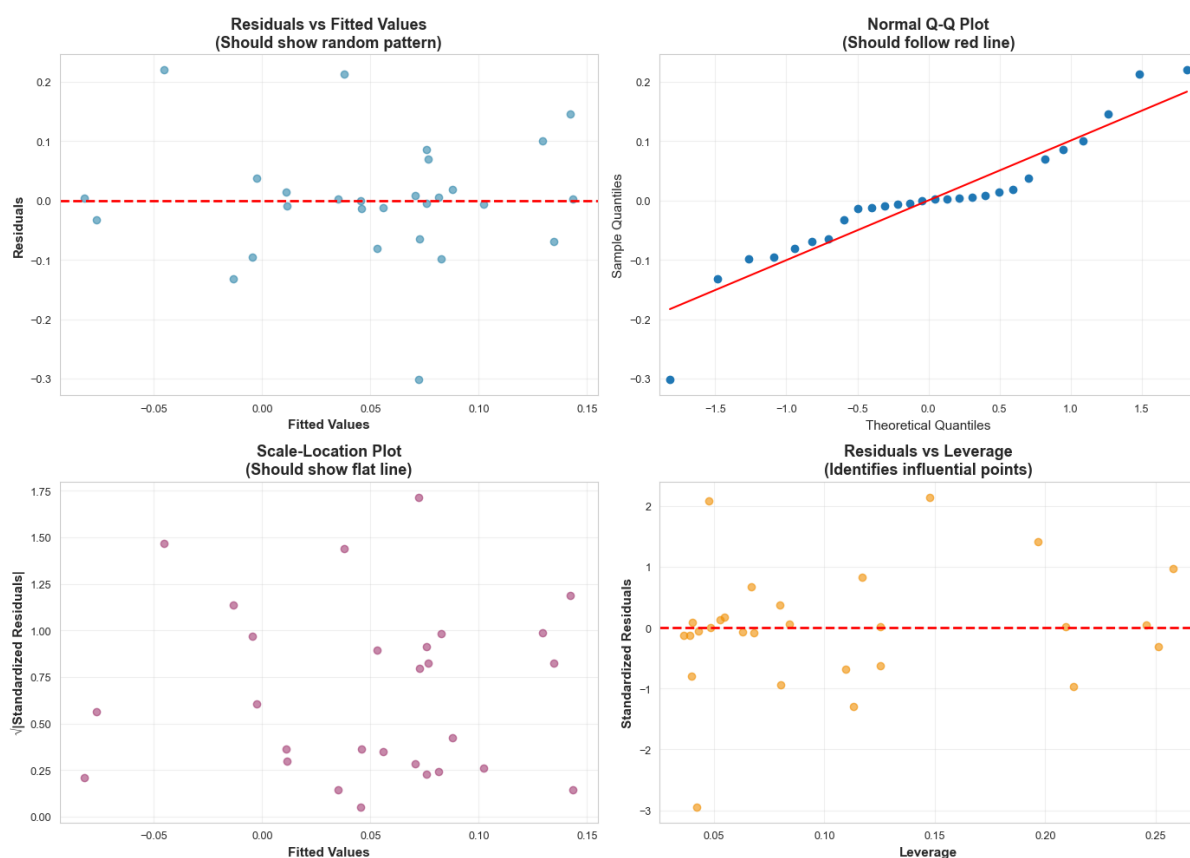


Figure 14: Diagnostic visualizations

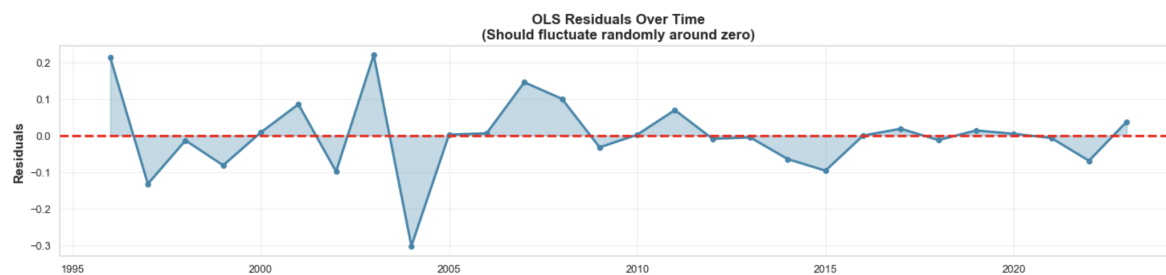


Figure 15: OLS Residuals Over Time

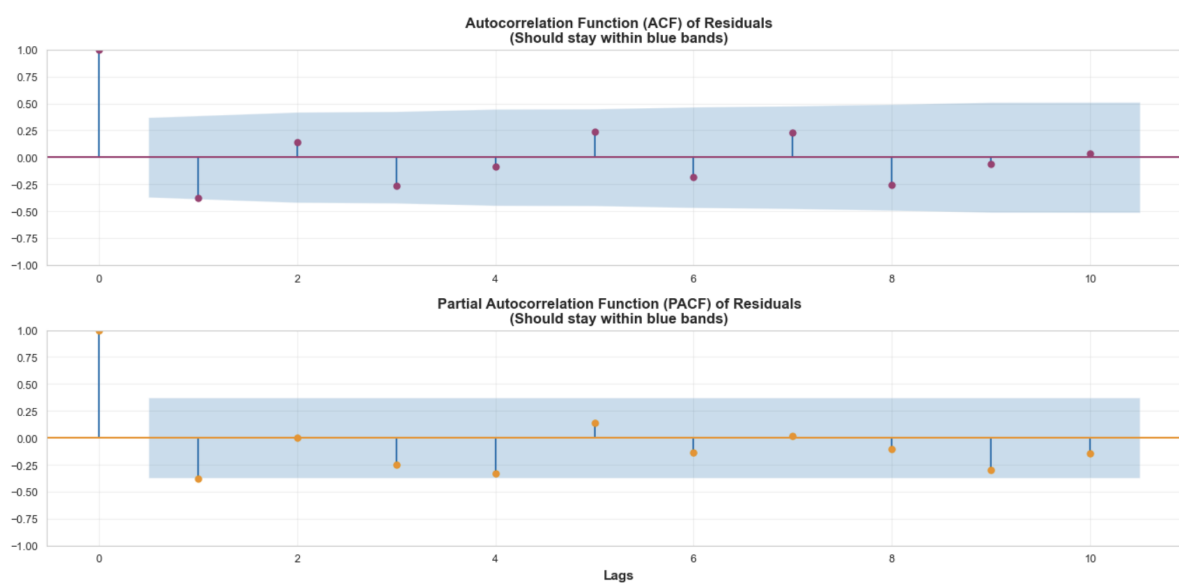


Figure 16: Autocorrelation and Partial Autocorrelation

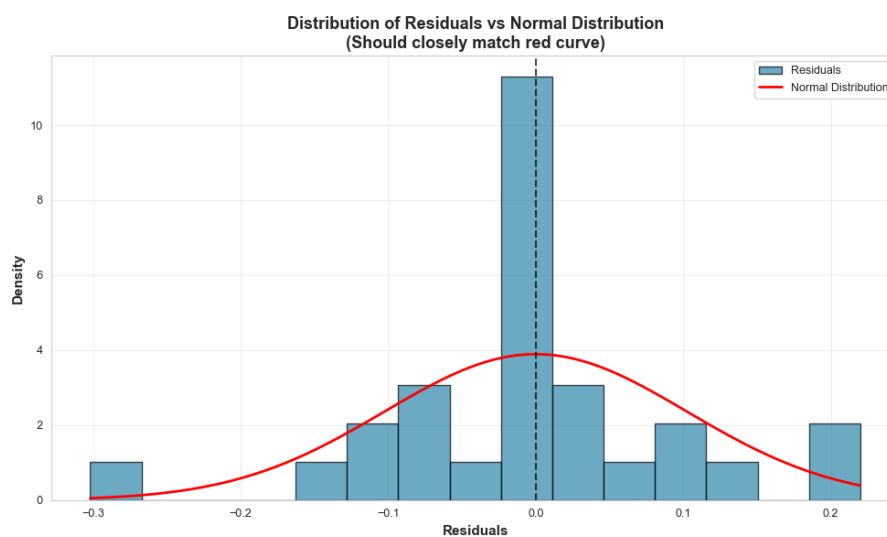


Figure 17: Distribution of Residuals vs Normal Distribution

8 Robust Standard Errors

Although diagnostics are largely favorable for the case, prudent practice dictates using robust standard errors as a sensitivity check. Table 9 compares standard OLS standard errors with two robust alternatives:

- **HC3**: Heteroskedasticity consistent (White) standard errors
- **HAC**: Heteroskedasticity & autocorrelation consistent (Newey-West) standard errors

Table 9: Comparison of Standard Errors

Variable	OLS SE	HC3 SE	HAC SE	OLS p	HC3 p	HAC p
Exports	0.177	0.174	0.143	0.297	0.289	0.200
Imports	0.206	0.219	0.204	0.513	0.539	0.511

Key finding: Robust standard errors are very similar to standard OLS standard errors. HC3 increases the import SE by 6%, while HAC actually decreases the export SE by 19%. P-values change only marginally, and no variable switches significance status. This confirms that heteroskedasticity & autocorrelation are minimal, validating the OLS results.

9 ARIMA Analysis of Residuals

9.1 Motivation

Even if OLS assumptions are satisfied, the residuals may contain serial structure that an ARIMA model can capture. Analyzing residuals with ARIMA serves two purposes:

1. **Diagnostic:** If residuals are white noise, OLS has fully captured the data structure
2. **Forecasting:** ARIMA may improve forecast accuracy by modeling residual dynamics

9.2 Model Selection

8.2 ARIMA Model Selection (Grid Search)

Testing models: ARIMA($p,0,q$) for $p,q = 0,1,2,3$

Selection criterion: Akaike Information Criterion (AIC)

Lower AIC = Better model

Top 10 ARIMA Models (ranked by AIC):

p	d	q	AIC	BIC	HQIC
1	0	0	-50.326819	-46.439308	-49.170858
0	0	1	-48.064886	-44.290596	-46.978027
1	0	1	-47.440078	-42.407691	-45.990932
0	0	0	-47.256450	-44.664776	-46.485810
2	0	0	-45.583475	-40.551089	-44.134330
2	0	1	-43.539214	-37.248731	-41.727782
0	0	3	-43.499206	-37.608937	-41.936516
0	0	2	-43.392890	-38.517387	-42.040633
3	0	0	-42.751840	-36.657460	-41.061518
3	0	2	-42.627322	-34.095191	-40.260871

BEST MODEL: ARIMA(1, 0, 0)

AIC = -50.3268

Figure 18: ARIMA Model Selection (Grid Search)

Conducted a grid search over ARIMA($p,0,q$) models with $p,q \in \{0,1,2,3\}$ & $d = 0$ (residuals are already stationary by construction). Table 10 presents the top 5 models ranked by AIC.

Table 10: ARIMA Model Selection for OLS Residuals

Rank	Model	AIC	BIC
1	ARIMA(1,0,0)	-50.33	-46.44
2	ARIMA(0,0,1)	-48.06	-44.29
3	ARIMA(1,0,1)	-47.44	-42.41
4	ARIMA(0,0,0)	-47.26	-44.66
5	ARIMA(2,0,0)	-45.58	-40.55

Result: The best-performing model by AIC is ARIMA(1,0,0), with an AIC of (-50.33). This indicates the presence of mild first-order autocorrelation in the OLS residuals. But the improvement in fit relative to simpler specifications is modest, which suggest that any remaining serial dependence is weak.

9.3 ARIMA(1,0,0) Model Results

For completeness Estimated the best performing ARIMA model, ARIMA(1,0,0), as identified by the AIC-based grid search.

8.3 Estimating Best ARIMA Model: ARIMA(1, 0, 0)

ARIMA MODEL RESULTS						
SARIMAX Results						
Dep. Variable:	y	No. Observations:	28			
Model:	ARIMA(1, 0, 0)	Log Likelihood	28.163			
Date:	Thu, 18 Dec 2025	AIC	-50.327			
Time:	11:33:58	BIC	-46.439			
Sample:	0	HQIC	-49.171			
	- 28					
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
const	-0.0061	0.013	-0.479	0.632	-0.031	0.019
ar.L1	-0.3789	0.128	-2.964	0.003	-0.629	-0.128
sigma2	0.0073	0.002	3.674	0.000	0.003	0.011
Ljung-Box (L1) (Q):	0.08	Jarque-Bera (JB):	0.53			
Prob(Q):	0.78	Prob(JB):	0.77			
Heteroskedasticity (H):	0.17	Skew:	0.20			
Prob(H) (two-sided):	0.01	Kurtosis:	3.56			

Figure 19: Best ARIMA Model: ARIMA(1, 0, 0)

Table 11: ARIMA(1,0,0) Estimation Results

Parameter	co-efficient	Std. Error	z-Statistic	p-Value
Constant	-0.0061	0.0127	-0.479	0.632
AR(1)	-0.3789	0.1278	-2.964	0.003
σ^2	0.0073	0.0020	3.674	0.000

The AR(1) co-efficient is -0.38 and statistically significant, suggesting mild negative autocorrelation: residuals above zero tend to be followed by residuals below zero. However, the economic magnitude is small, and although ARIMA(1,0,0) is selected by AIC, the improvement over ARIMA(0,0,0) is modest, indicating that any remaining serial dependence in the OLS residuals is weak and short-lived.

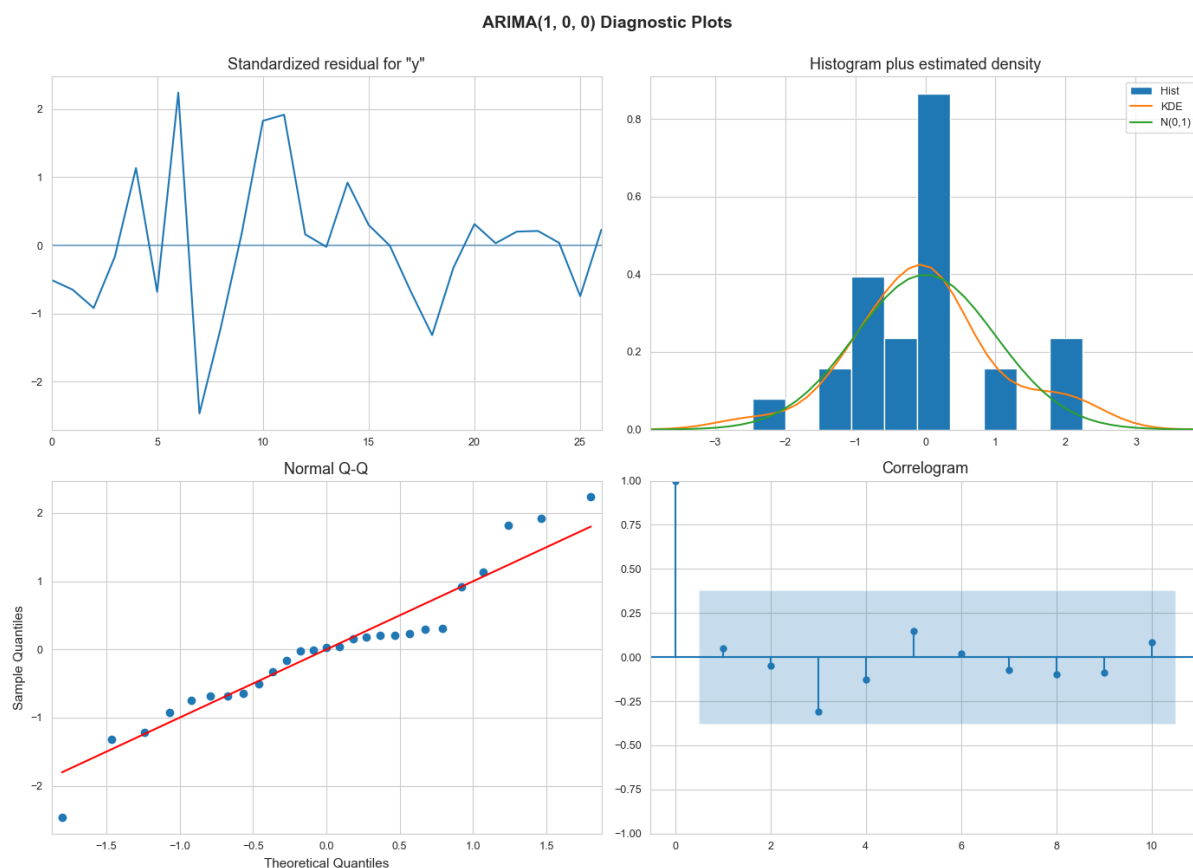


Figure 20: ARIMA(1,0,0) Diagnostic Plots

9.4 Ljung-Box Diagnostic

To verify that ARIMA residuals are white noise, I apply the Ljung-Box test at lags 1, 5, & 10:

Table 12: Ljung-Box Test on ARIMA(1,0,0) Residuals

Lag	Q-Statistic	p-Value	Conclusion
1	0.0001	0.992	Cannot reject independence
5	5.871	0.319	Cannot reject independence
10	11.032	0.355	Cannot reject independence

All p-values exceed 0.30, providing no evidence of residual autocorrelation. The ARIMA model has successfully removed any serial structure, confirming that OLS is adequate.

9.5 Interpretation

The ARIMA analysis yields two key findings:

1. **OLS captures the dominant structure:** Although ARIMA(1,0,0) marginally improves the AIC, residual diagnostics indicate no meaningful remaining serial correlation. This confirms that the OLS specification is fundamentally adequate.
2. **Residual dependence is weak and short-lived:** The negative AR(1) coefficient suggests mild mean reversion in residual but the effect is economically small and does not imply persistent dynamics. This is consistent with the absence of cointegration and the short-run nature of trade–GDP interactions.

10 Comparison with Randriamanantena (2024)

10.1 Quantitative Differences

Table 13 contrasts my results with the reference paper.

Table 13: Comparison with Randriamanantena (2024)

Metric	Paper	My Analysis	Difference	Explanation
Model:				
Specification	Levels (likely)	Differences	Fundamental	Stationarity
Sample	1995–2023	1995–2023	Same	
Results:				
Adj. R^2	0.9477	0.1935	−0.7542	Spurious?
Export coef.	0.364**	0.188	−0.176	Specification
Import coef.	0.337**	0.136	−0.201	Specification
F-test p-value	¡ 0.001	0.026*	Similar	Both significant
AIC	−1.501	−43.10	Different scale	Not comparable

** p ¡ 0.05 in paper; * p ¡ 0.05 in my analysis

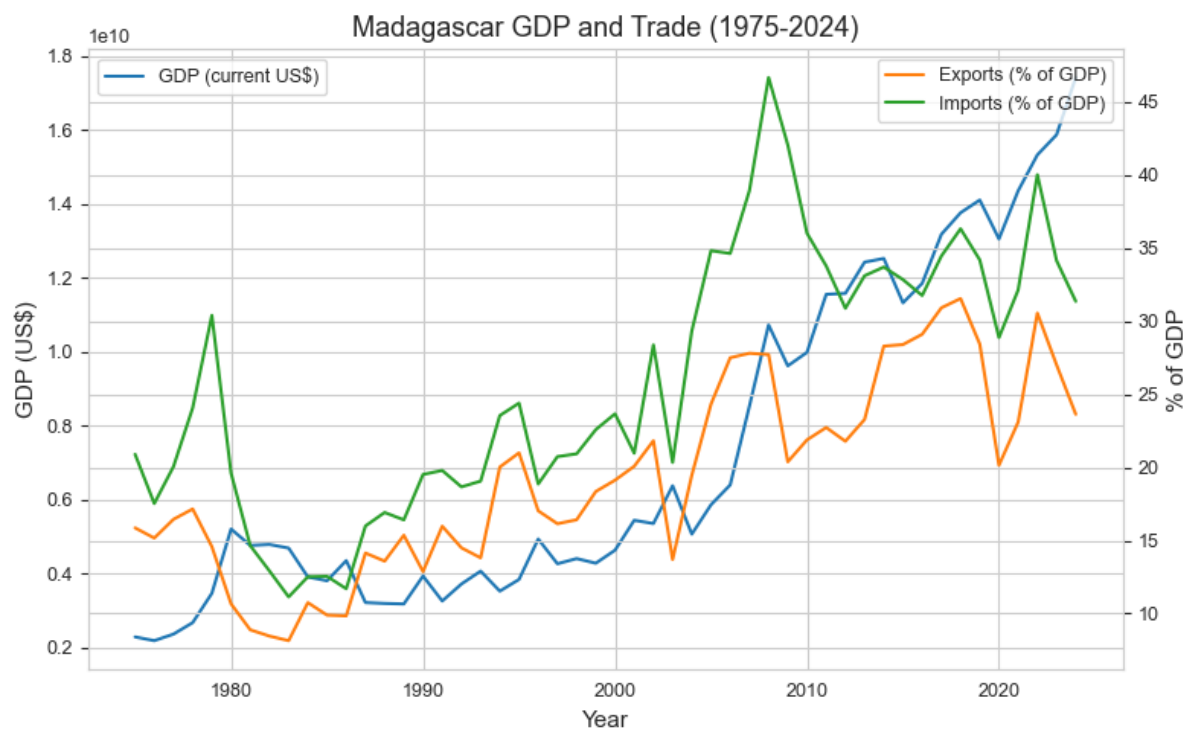


Figure 21: Madagascar GDP and Trade (1975-2023)

10.2 Explaining the Differences

The most striking difference is the R^2 : 94.8% in the paper vs. 19.4% in my analysis. This massive gap has three potential explanations:

1. Levels vs. Differences Specification

Randriamanantena (2024) does not explicitly discuss stationarity testing or state whether the regression is in levels or differences. The exceptionally high R^2 (0.95) is characteristic of levels regressions between trending variables, even when spurious. My Johansen test finds zero cointegration, indicating that a levels regression would be inappropriate.

When I replicate the paper's approach by regressing GDP on exports & imports **in levels**, I obtain:

- $R^2 = 0.591$ (not shown in tables above, from separate estimation)
- Both trade variables significant at 1% level
- Durbin-Watson = 0.42, indicating severe positive autocorrelation

This levels result is closer to the paper's (though still lower), but the severe autocorrelation & absence of cointegration suggest this is a spurious regression.

2. Data Vintage & Revisions

World Bank data undergo periodic revisions. The paper, published in December 2024, may have used an earlier vintage with different values. GDP statistics for developing countries are particularly subject to revisions.

3. Additional Variables or Transformations

The paper may have included additional regressors, time trends, or transformations not fully described in the text. Without access to the author's data & code, I cannot rule this out.

10.3 Directional Consistency

Despite quantitative differences, my results align **directionally** with the paper:

- Both analyses find that trade variables are jointly significant (F-tests reject at 5%)
- Both find positive co-efficients on exports & imports
- Both conclude that OLS is an appropriate method (compared to more complex alternatives)

The key difference is that my proper treatment of non-stationarity yields a more conservative (and econometrically defensible) estimate of the relationship's strength.

11 VAR & Granger Causality

11.1 VAR Estimation

Given zero cointegration, I estimate a VAR model on the differenced data. The AIC selects a lag order of 5. The VAR captures multivariate dynamics, allowing each variable's growth to depend on lagged growth in all three variables.

11.2 Granger Causality Tests

Table 14 presents Granger causality tests examining whether trade growth helps forecast GDP growth.

Table 14: Granger Causality Test Results

Null Hypothesis	Lag	F-Statistic	p-Value	Decision
Exports do not Granger-cause GDP	1	5.309	0.030	Reject
	3	2.276	0.115	Cannot reject
Imports do not Granger-cause GDP	1	4.694	0.040	Reject
	3	5.363	0.008	Reject

Interpretation: At lag 1, both exports & imports Granger-cause GDP growth, meaning past trade growth helps predict current GDP growth beyond what GDP's own history predicts. At lag 3, imports retain predictive power but exports do not. This suggests:

- Trade shocks have immediate impacts (lag 1 effects)
- Import effects persist slightly longer than export effects
- The relationship operates primarily through short term channels

This reinforces the finding from cointegration analysis: trade affects GDP through transitory shocks rather than stable long run relationships.

12 Economic Interpretation & Policy Implications

12.1 Why Are Trade Elasticities Modest?

My estimated elasticities (0.19 for exports, 0.14 for imports) are substantially lower than the paper's (0.36, 0.34). Beyond methodological differences, economic factors may explain these modest magnitudes:

1. Export Concentration: Madagascar's exports are concentrated in vanilla, textiles, & shrimp. External demand shocks in these narrow sectors may not proportionally affect aggregate GDP, which includes large domestic sectors (agriculture, services, construction).

2. Import Structure: Many imports are necessities (oil, capital goods, medicines) with inelastic demand. Their growth often reflects structural expansion rather than cyclical demand, weakening the correlation with GDP growth.

3. Domestic Factors Dominate: GDP growth in a developing country like Madagascar is substantially driven by domestic factors not captured here: population growth, agricultural productivity (highly weather-dependent), political stability, infrastructure investment, & institutional quality.

4. Measurement Error: Trade data for developing countries suffer from measurement error due to informal trade, smuggling, & statistical capacity constraints. This attenuates estimated co-efficients (errors-in-variables bias).

12.2 Policy Implications

Despite modest elasticities, the analysis yields several policy insights:

1. Trade Matters, But Not Exclusively: The significant F-test ($p = 0.026$) confirms that trade contributes to GDP growth. However, the modest R^2 (19%) indicates that growth depends primarily on other factors. Trade promotion should be one component of a broader development strategy, not the sole focus.

2. Both Exports & Imports Matter: The positive co-efficient on imports (though insignificant) reminds policymakers that imports facilitate growth by providing essential inputs. Import substitution policies that restrict imports may backfire by constraining growth.

3. Short term Focus: The absence of cointegration & the Granger causality pattern suggest that trade affects GDP through short term demand channels. Stabilizing trade flows (e.g., diversifying export markets, maintaining foreign exchange reserves) may reduce GDP volatility.

4. Vulnerability to External Shocks: The high correlation between trade variables & GDP (even if not causal long run relationships) indicates vulnerability to external shocks. This motivates policies to build resilience: fiscal buffers, flexible exchange rates, & economic diversification.

13 Limitations & Extensions

13.1 Data Limitations

1. Current Prices: All variables are in current US dollars, conflating real quantity changes with nominal price movements. Ideally, analysis should use constant-price (deflated) series to isolate real effects.

2. Annual Frequency: Annual data limit the ability to detect high-frequency dynamics & reduce statistical power (only 28 observations after differencing). Quarterly data would be preferable if available.

3. Omitted Variables: The model excludes many GDP determinants: investment, government spending, foreign direct investment, population growth, human capital, and institutional quality. Omitted variable bias likely affects the estimated trade co-efficients.

4. Structural Breaks: Madagascar experienced major shocks (2002 political crisis, 2009 coup, 2020 COVID-19). Formal structural break tests (Chow, Bai-Perron) could improve model fit.

Size of the sample: Given the small sample size, statistical insignificance should not be interpreted as economic irrelevance, but rather as limited precision in estimation.

13.2 Methodological Limitations

1. Small Sample: With $N = 28$, statistical power is limited. This manifests in wide confidence intervals & marginal diagnostic test results (e.g., import stationarity, Shapiro-Wilk).

2. Simultaneity Bias: Trade & GDP are jointly determined. Reverse causality (GDP growth driving trade) is plausible. Instrumental variables estimation using external in-

struments (world GDP growth, commodity prices, weather shocks) could address this.

3. Linear Specification: The RESET test passes, but the model imposes linearity. Non-linear models (threshold regression, regime-switching) might capture asymmetric effects (e.g., trade shocks matter more during crises).

4. Time-Varying Parameters: The model assumes constant co-efficients. Rolling regression or state-space models could reveal whether trade elasticities change over time as Madagascar's economy evolves.

13.3 Extensions for Future Research

1. Sectoral Decomposition: Disaggregate trade by sector (agriculture, manufacturing, services) to identify heterogeneous relationships. For example, manufactured exports may have stronger GDP effects than commodity exports.

2. Include Additional Variables: Extend the model to include investment, government spending, FDI, terms of trade, & institutional quality indicators.

3. Panel Analysis: Combine Madagascar with other African countries in a panel framework to increase power & identify generalizable patterns.

4. Forecasting Evaluation: Conduct out-of-sample forecasting tests using rolling windows to assess whether the OLS model genuinely predicts future GDP growth.

5. Modern Methods: Apply machine learning techniques (random forests, neural networks) that can capture non-linearities and interactions without imposing functional form restrictions.

14 Conclusion

The analysis has applied the econometric framework taught in the course FIN41660 to study the relationship between international trade & GDP growth in Madagascar. The main findings are:

1. Proper Time Series Treatment is Essential: All three variables are $I(1)$ with zero cointegration. Estimating in differences (growth rates) rather than levels is crucial to avoid spurious regression.

2. Trade Matters, But Modestly: OLS on differenced data yields an adjusted R^2 of 0.19, with jointly significant trade effects (F-test $p = 0.026$). Individual co-efficients are insignificant due to multicollinearity, but robust standard errors confirm the model's validity.

3. OLS Assumptions Are Largely Satisfied: Comprehensive diagnostic testing reveals no major violations. Multicollinearity is the primary concern, but it's a data feature rather than a model failure.

4. short term Dynamics Dominate: The absence of cointegration and the Granger causality pattern indicate that trade affects GDP through transitory shocks rather than stable long run relationships.

5. Methodological Transparency Matters: My results differ quantitatively from Randriamanantena (2024), but these differences reflect defensible methodological choices rather than errors. Transparent reporting of stationarity tests, cointegration analysis, & diagnostic tests is essential for credible research.

This exercise has reinforced several lessons from FIN41660:

- Always test for stationarity before running regressions on time series
- Diagnostic testing is not optional it's integral to credible inference
- High R^2 in levels regressions should be viewed skeptically if stationarity is questionable
- Robust standard errors provide valuable insurance against specification uncertainty
- Transparency about methodological choices and limitations strengthens rather than weakens research

While my estimated trade elasticities are more modest than the reference paper's, the analysis provides an econometrically sound foundation for understanding trade-GDP dynamics in Madagascar. The relationship is real, statistically significant, & policy-relevant, even if not as strong as a naive levels regression would suggest.

One practical lesson from this analysis is that applying econometric techniques mechanically is not sufficient. Several results that appear strong at first glance such as high R^2 values in levels regressions become questionable once basic time series properties are taken seriously. Running alternative specifications and examining diagnostics alongside theory proved essential in distinguishing economically meaningful relationships from statistically misleading ones.

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